

Patient Details

Name : Mr. ASIM KUMAR BANERJEE
OQG2604130367 & Mrs.
BRATATI BANERJEE
OQG2604130370

Sex / Age : Male / 71 Years; Female / 64
Years

Case ID : 60400121681/ 60400121684

Ref By :

PT. ID : 60400121681/ 60400121684

Test Name : ORION (V2) (WES-Whole
Exome Sequencing)

Bill. Loc. : Oncquest Laboratories Ltd.

Sample Details

Registration Date & Time : 2026-04-13
15:16:38

Sample Type : Whole Blood EDTA

Sample Date & Time : 2026-04-13
15:16:00

Ref ID 1. : -

Report Date & Time : 2026-05-09
10:25:14

PT. Loc. : -

Summary of the Clinical History Provided

Partner 1 (Name: Mrs. BRATATI BANERJEE, NCGM ID: 60400121684) and Partner 2 (Name: Mr. ASIM KUMAR BANERJEE, NCGM ID: 60400121681) are a couple referred for carrier screening in view of the proband history as follows:

Salient features: Primary aceruloplasminemia, Iron overload (hepatic + mixed; Kupffer cells > hepatocytes), No DM, No iron deposition in brain.

Investigations / Tests performed: Ferritin 1160 ng/ml, INR 1.04, LFT results varied, ANA Positive, ASMA Positive, Total IgG Increased, HFE gene mutations (H63D, S65C, C282Y) not detected.

Clinical suspicion: Primary aceruloplasminemia, ? AIH (autoimmune hepatitis)

Investigations: WES Proband - Dr. RUPSA BANERJEE OQG2603120880 (NCGM ID: 60300113628)

Probable compound heterozygous variants in CP gene [(c.583G>A; p.Gly195Arg), (c.382A>G; p.Lys128Glu)] associated with Aceruloplasminemia (AR).

Trio analysis have been performed among Mrs. BRATATI BANERJEE (NCGM ID: 60400121684) Mr. ASIM KUMAR BANERJEE (NCGM ID: 60400121681), and proband Dr. RUPSA BANERJEE (NCGM ID: 60300113628).

Test Results and Interpretation

PREVIOUSLY DETECTED CP GENE VARIANTS [(c.583G>A; p.Gly195Arg), (c.382A>G; p.Lys128Glu)] IN PROBAND, SEGREGATED IN BOTH PARTNERS RESPECTIVELY.

UPON TRIO ANALYSIS, NO OTHER SIGNIFICANT DE NOVO/ OTHER SIGNIFICANT VARIANTS DETECTED.

Summary of Couple Screening

Gene and Transcript	Variant	BRATATI BANERJEE OQG2604130370 (Partner 1)	ASIM KUMAR BANERJEE OQG2604130367 (Partner 2)	Interpretation
CP (NM_000096.4)	Nomenclature: c.583G>A p.Gly195Arg Involved /Total Exon-Intron Exon 3/Exon 19 Classification: Pathogenic Diseases: Aceruloplasminemia Inheritance: Autosomal recessive	Present (Heterozygous) 46.938	Absent -	Present (Compound Heterozygous)
	Nomenclature: c.382A>G p.Lys128Glu Involved /Total Exon-Intron Exon 2/Exon 19 Classification: Uncertain significance Diseases: Aceruloplasminemia Inheritance: Autosomal recessive	Absent -	Present (Heterozygous) 55.629	

Details of variant

This section describes the variant and disease associated with the gene correlating/matching with the given phenotype.

CP (NM_000096.4) - [c.583G>A; p.Gly195Arg]

Type of variant	gnomAD frequency	Computational evidences	ClinVar	Null Variant	Downstream Null Variant	Previously reported [reported zygosity]	Variant references
Missense Variant	0.0006%	REVEL: 0.972	Likely Pathogenic	NA	NA	Yes [Homozygous]	Kono S et al., 2006

This missense change has been observed in individual(s) with aceruloplasminemia (Kono S et al., 2006). Experimental studies have shown that this missense change affects CP function (Kono S et al.,

Details of variant

2006, Thepsuwan P et al., 2023). Based on the available evidences this variant is classified as pathogenic.

References:

1. Kono S et al., Molecular and pathological basis of aceruloplasminemia. Biol Res. 2006;39(1):15-23.
2. Kono S et al., Biochemical features of ceruloplasmin gene mutations linked to aceruloplasminemia. Neuromolecular Med. 2006;8(3):361-74.
3. Thepsuwan P et al., Hepatic SELIL-HRD1 ER-associated degradation regulates systemic iron homeostasis via ceruloplasmin. Proc Natl Acad Sci U S A. 2023 Jan 10;120(2):e2212644120.

CP (NM_000096.4) - [c.382A>G; p.Lys128Glu]

Type of variant	gnomAD frequency	Computational evidences	ClinVar	Null Variant	Downstream Null Variant	Previously reported [reported zygosity]	Variant references
Missense Variant	0.0000%	REVEL: 0.963	Not submitted	NA	NA	No	NA

Disease: ACERULOPLASMINEMIA [OMIM GENE ID * 117700]

Aceruloplasminemia is characterized by iron accumulation in the brain and viscera. The clinical triad of retinal degeneration, diabetes mellitus (DM), and neurologic disease is seen in individuals ranging from age 30 years to older than 70 years. The neurologic findings of movement disorder (blepharospasm, grimacing, facial and neck dystonia, tremors, chorea) and ataxia (gait ataxia, dysarthria) correspond to regions of iron deposition in the brain. Individuals with aceruloplasminemia often present with anemia prior to onset of DM or obvious neurologic problems. Cognitive dysfunction including apathy and forgetfulness occurs in more than half of individuals with this condition.

Special Recommendations / Notes

- Only pathogenic and likely pathogenic variants with respect to current scientific evidence are reported in non-phenotype related genes. variants of uncertain significance are analyzed in the respective spouse and their status is provided in non phenotype related genes.
- Variant classification is known to change with time and emerging evidence. Follow up with respect to changes in classification with your referring clinician/laboratory is recommended at 3-5 yearly intervals or as deemed required by your clinician.
- The clinical history section in the report contains concised details of the salient features provided for the patient. The analysis has been performed with all the clinical details provided and relevant HPO terms based on the same.

Test Information

1. Clinical correlation as well as reverse phenotyping is recommended for all reports.
2. Genetic counseling for accurate interpretation of test results is recommended.
3. The reported findings are based on NGS analysis.
4. Analysis includes both single nucleotide (SNV) as well as copy number variant analysis (CNV).
5. Copy number variants when detected are included in the report.

Test Information

6. Since CNV analysis is performed on a comparative basis, a negative result does not exclude the presence of a CNV.
7. The CNV pipeline is not validated for <3 exon copy number variants wherein detection is influenced by the underlying gene region and structure.
8. Variant calling (SNV and CNV) may be limited in low covered regions as well as in regions of low complexity and in pseudogenes.
9. Synonymous variants (not affecting splice site) as well as intronic variants are usually not reported.
10. Analysis and reporting is focussed on the provided phenotype and based on relevant HPO (Human Phenotype Ontology) terms as well as on genes associated with provided phenotype.
11. A genotype based analysis is also performed when the above yields negative results but reporting is limited to genes wherein current available evidence suggests a possible association with the provided phenotype.
12. It may not be possible to fully resolve certain details about variants, such as mosaicism, phasing, or mapping ambiguity.
13. Disease descriptions are included from OMIM, Genereviews and PUBMED indexed articles as and where applicable.
14. The test methodology currently does not detect large deletions/duplications, triplet repeat expansions and epigenetic changes. The test also does not include analysis of predictors for multifactorial, polygenic and/or complex diseases.
15. Phenotype variability may be due to modifying genetic/non-genetic factors and is not a part of the current analysis.
16. Candidate genes and genes with limited evidence are designated as genes of uncertain significance and all variants detected therein are classified as variants of uncertain significance.
17. Typically only variants at a depth of $\geq 10X$ are reported. Lower depth variants may be false positives.
18. Detected variants in low complexity regions as well as variants at low depth and relatively low VAF should be reconfirmed by an alternate methodology.
19. CNV confirmation via MLPA or Exon array is recommended for copy number variants involving a single gene as well as for CNV <400kb. Only large constitutional CNV can be tested via other array platforms.
20. Variant depth/Total depth has been mentioned in summary of the variants.
21. Parental/Maternal testing as applicable is recommended for variants when detected for phasing (where applicable)
22. Segregation analysis (testing of multiple affected as well as unaffected members) of detected variants (if any) is recommended. Variant classification is subject to change after segregation.
23. ACMG secondary findings as well as carrier status of variants are only provided when requested.
24. Within carrier screening risk factors/alleles as well as hypomorphic variants may not be included
25. Interobserver as well as inter-laboratory variation is known with respect to variant classification due to the subjective nature of the provided criteria. Though the laboratory follows the updated recommendations provided by the ClinGen SVI as well as ACMG, independent assessment of variant classification by the referring clinician is recommended before decision making.
26. For prenatal samples analysis is limited to provided clinical phenotypes utilizing relevant HPO terms. Typically variants of uncertain significance are only introduced if the respective gene has been associated with the observed fetal phenotype. In solo fetal exomes, variants of uncertain deemed to be disease causing based on genotype characteristics may be included for further evaluation by the referring clinician. In prenatal scenarios trio fetal testing is strongly recommended to allow better interpretation of detected variants.
27. If the above results do not correlate completely with patient phenotype, additional testing is advised based on clinician's discretion.
28. Typically, heterozygous variants of uncertain significance in genes associated with autosomal recessive disorder are not reported in the proband. Such variants are included if relevant to phenotype in carrier screening.
29. A negative report does not exclude a genetic disorder due to inherent limitations of the assay design.
30. On the background of whole exome sequencing, analysis is limited to provided indications/ requested testing. Hence analysis is limited to a single gene if the same has been requested.
31. As a part of knowledge sharing initiative, all reported variants are submitted in de-identified form in the ClinVar database.
32. Extracted DNA if available after requisitioned testing will be stored as per recommendations. Please note that DNA may degrade over time and this may affect the quality of the stored sample.
33. Collected blood samples are not stored
34. Prenatal samples (AF/ CVS/cord blood) are not stored. Extracted DNA if available is stored as acknowledged above.
35. As per PCPNDT fetal gender is not revealed.
36. Maternal cell contamination is recommended for prenatal and POC (product of conception) to ensure accuracy of test results. The same is performed only when requested and on the availability of the maternal sample.
37. Discrepant maternal cell contamination results may rise with use of donor gamete and hence information regarding the same should be provided to the laboratory.
38. Reproductive decision making is not recommended based on variants of uncertain significance.

Test Information

39. Raw data can be transferred on request and after due consent/assent from the involved patient/ family. Additional charges will be applicable for the same.
40. The test performed by the laboratory with the assumption that the sample belongs to the person herewith mentioned in the requisition form and appropriate consent as well as prenatal counseling including test information has been provided by the referring clinician.
41. Repeat sampling may be required in case of gender discrepancy (unless the same can be attributed to an underlying scientific reason) as well as in rare cases where DNA/data quality prevents further analysis.
42. Reanalysis of data is recommended as deemed necessary by the referring clinician. Additional charges will be applicable for the same.

Technical Notes

Methodology – Massively Parallel Sequencing (Next Generation Sequencing): Genomic DNA from the submitted specimen was enriched for the complete coding regions and splice site junctions of genes listed below using a custom bait- capture system. Paired End Sequencing was performed with 2x100/2x150 chemistry. Reads were assembled and were aligned to reference sequences based on NCBI RefSeq transcripts and human genome build GRCh37/UCSC hg19. Data was filtered and analyzed to identify variants of interest and interpreted in the context of a single most damaging, clinically relevant transcript for the purpose of the report, indicated as a part of variant details. Enrichment and analysis focus on the coding sequence of the indicated transcripts, 5-10bp of flanking intronic sequence, and other specific genomic regions demonstrated to be causative of disease at the time of assay design. Deletion and duplication analysis is performed in cases when indicated but detected variations need to be confirmed by an alternate methodology. Sequence and copy number variants are reported according to the Human Genome Variation Society (HGVS).

Laboratory reporting protocol: The analysis is based on the provided phenotype: relevant HPO terms, curated gene panels and relevant literature is assessed for phenotype based analysis. Variant reporting is limited to exon regions and upto 10 basepairs within exon intron boundaries. Previously reported deep intronic and non coding variants will be included when detected at a depth more than 10X . Variant reporting is performed at a minimum depth of 10X. The gnomAD variant frequency reflects the liftover of hg38 to hg19.

For Mitochondrial Genome Sequencing (if requested): Only phenotype-related Pathogenic and Likely Pathogenic variations reported in the MitoMap database as well as literature are reported. Haplogroups are not analyzed. A list of variants other than the above are available on request. Analyzed genes include:MT-ND1, MT-ND2, MT-ND3, MT-ND4L, MT-ND4, MT-ND5, MT-ND6, MT-CYB, MT-CO1, MT-CO2, MT-CO3, MT-ATP6, MT-ATP8, MT-RNR2, MT-RNR1, MT-RNR2, MT-TA, MT-TR, MT-TN, MT-TD, MT-TC, MT-TE, MT-TQ, MT-TG, MT-TH, MT-TI, MT-TL1, MT-TL2, MT-TK, MT-TM, MT-TF, MT-TP, MT-TS1, MT-TS2, MT-TT, MT-TW, MT-TY, MT-TV.

Tools and Databases employed for analysis: Clinvar, OMIM, HGMD, UCSC genome browser, Uniprot, Ensembl, dbSNP, gnomAD, ExAC, Pubmed, Dgap, icgc, Kaviar, various bioinformatics analysis, predictive tools and disease specific databases used as available and appropriate. Such tools/databases would be mentioned wherever used.

REVEL: The REVEL score for an individual missense variant can range from 0 to 1, with higher scores reflecting greater likelihood that the variant is disease-causing.

Bioinformatics pipeline version: v16.1.1

Gene Coverage

Indication Based Analysis:

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
A4GALT	P-1: 100 % P-2: 100 %	AARS2	P-1: 100 % P-2: 100 %	ABAT	P-1: 100 % P-2: 100 %	ABCB11	P-1: 100 % P-2: 100 %
ABCB4	P-1: 100 %	ABCB6	P-1: 100 %	ABCB7	P-1: 100 %	ABCC2	P-1: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
	P-2: 100 %		P-2: 100 %		P-2: 100 %		P-2: 100 %
ABCD1	P-1: 100 % P-2: 100 %	ABCD3	P-1: 100 % P-2: 100 %	ACAD9	P-1: 100 % P-2: 100 %	ACADM	P-1: 100 % P-2: 100 %
ACADS	P-1: 100 % P-2: 100 %	ACADSB	P-1: 100 % P-2: 100 %	ACADVL	P-1: 100 % P-2: 100 %	ACAT1	P-1: 96.3 % P-2: 97.2 %
ACBD5	P-1: 100 % P-2: 100 %	ACD	P-1: 100 % P-2: 100 %	ACO2	P-1: 100 % P-2: 100 %	ACOD1	P-1: 100 % P-2: 100 %
ACOX1	P-1: 100 % P-2: 100 %	ACOX2	P-1: 100 % P-2: 100 %	ADK	P-1: 100 % P-2: 100 %	AFG3L2	P-1: 100 % P-2: 100 %
AGA	P-1: 100 % P-2: 100 %	AGK	P-1: 100 % P-2: 100 %	AGL	P-1: 100 % P-2: 100 %	AGPS	P-1: 100 % P-2: 100 %
AGXT	P-1: 100 % P-2: 100 %	AIFM1	P-1: 100 % P-2: 100 %	AKRID1	P-1: 100 % P-2: 100 %	ALAS2	P-1: 100 % P-2: 100 %
ALDH18A1	P-1: 100 % P-2: 100 %	ALDOA	P-1: 100 % P-2: 100 %	ALDOB	P-1: 100 % P-2: 100 %	ALG1	P-1: 100 % P-2: 100 %
ALG11	P-1: 100 % P-2: 100 %	ALG12	P-1: 100 % P-2: 100 %	ALG13	P-1: 100 % P-2: 100 %	ALG14	P-1: 100 % P-2: 100 %
ALG2	P-1: 100 % P-2: 100 %	ALG3	P-1: 100 % P-2: 100 %	ALG6	P-1: 100 % P-2: 100 %	ALG8	P-1: 95.4 % P-2: 95.4 %
ALG9	P-1: 99.9 % P-2: 99.9 %	AMACR	P-1: 100 % P-2: 100 %	ANO10	P-1: 100 % P-2: 100 %	APIB1	P-1: 100 % P-2: 100 %
AP1S1	P-1: 100 % P-2: 100 %	APTX	P-1: 100 % P-2: 100 %	ARCNI	P-1: 100 % P-2: 100 %	ARSB	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
ASAH1	P-1: 100 % P-2: 100 %	ATAD3A	P-1: 100 % P-2: 100 %	ATP13A2	P-1: 100 % P-2: 100 %	ATP5F1A	P-1: 100 % P-2: 100 %
ATP5F1D	P-1: 100 % P-2: 100 %	ATP5F1E	P-1: 100 % P-2: 100 %	ATP6AP1	P-1: 100 % P-2: 100 %	ATP6AP2	P-1: 100 % P-2: 100 %
ATP6VOA2	P-1: 100 % P-2: 100 %	ATP6V1A	P-1: 100 % P-2: 100 %	ATP6V1E1	P-1: 100 % P-2: 100 %	ATP7A	P-1: 100 % P-2: 100 %
ATP7B	P-1: 100 % P-2: 100 %	ATP8B1	P-1: 100 % P-2: 100 %	ATPAF2	P-1: 100 % P-2: 100 %	B3GALNT2	P-1: 100 % P-2: 100 %
B3GALT6	P-1: 100 % P-2: 100 %	B3GAT3	P-1: 100 % P-2: 100 %	B3GLCT	P-1: 100 % P-2: 100 %	B4GALNT1	P-1: 100 % P-2: 100 %
B4GALT1	P-1: 100 % P-2: 100 %	B4GALT7	P-1: 100 % P-2: 100 %	B4GAT1	P-1: 100 % P-2: 100 %	BAAT	P-1: 100 % P-2: 100 %
BCS1L	P-1: 100 % P-2: 100 %	BOLA3	P-1: 100 % P-2: 100 %	BTBD	P-1: 100 % P-2: 100 %	BVES	P-1: 100 % P-2: 100 %
C19orf12	P-1: 100 % P-2: 100 %	C1QBP	P-1: 100 % P-2: 100 %	CA5A	P-1: 100 % P-2: 100 %	CAD	P-1: 100 % P-2: 100 %
CANT1	P-1: 100 % P-2: 100 %	CARS2	P-1: 100 % P-2: 100 %	CAT	P-1: 100 % P-2: 100 %	CCDC115	P-1: 100 % P-2: 100 %
CCNI	P-1: 100 % P-2: 100 %	CCS	P-1: 100 % P-2: 100 %	CEL	P-1: 100 % P-2: 98.6 %	CEP89	P-1: 98.8 % P-2: 98.8 %
CFH	P-1: 100 % P-2: 100 %	CFTR	P-1: 100 % P-2: 100 %	CHCHD10	P-1: 100 % P-2: 100 %	CHCHD2	P-1: 100 % P-2: 100 %
CHIT1	P-1: 100 % P-2: 100 %	CHKB	P-1: 100 % P-2: 100 %	CHST14	P-1: 100 % P-2: 100 %	CHST3	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
CHST6	P-1: 100 % P-2: 100 %	CHSY1	P-1: 100 % P-2: 100 %	CISD2	P-1: 100 % P-2: 100 %	CLDN1	P-1: 100 % P-2: 100 %
CLN3	P-1: 100 % P-2: 100 %	CLN5	P-1: 100 % P-2: 100 %	CLN6	P-1: 100 % P-2: 100 %	CLN8	P-1: 100 % P-2: 100 %
CLPB	P-1: 100 % P-2: 100 %	CLPP	P-1: 100 % P-2: 100 %	CLPX	P-1: 100 % P-2: 100 %	COA3	P-1: 100 % P-2: 100 %
COA5	P-1: 100 % P-2: 100 %	COA6	P-1: 100 % P-2: 100 %	COA7	P-1: 100 % P-2: 100 %	COA8	P-1: 100 % P-2: 100 %
COASY	P-1: 100 % P-2: 100 %	COG1	P-1: 100 % P-2: 100 %	COG2	P-1: 100 % P-2: 100 %	COG4	P-1: 100 % P-2: 100 %
COG5	P-1: 100 % P-2: 100 %	COG6	P-1: 100 % P-2: 100 %	COG7	P-1: 100 % P-2: 100 %	COG8	P-1: 100 % P-2: 100 %
COPA	P-1: 100 % P-2: 100 %	COPB2	P-1: 100 % P-2: 100 %	COQ2	P-1: 100 % P-2: 100 %	COQ4	P-1: 100 % P-2: 100 %
COQ5	P-1: 100 % P-2: 100 %	COQ6	P-1: 100 % P-2: 100 %	COQ7	P-1: 100 % P-2: 100 %	COQ8A	P-1: 100 % P-2: 100 %
COQ8B	P-1: 100 % P-2: 100 %	COQ9	P-1: 100 % P-2: 100 %	COX10	P-1: 100 % P-2: 100 %	COX14	P-1: 100 % P-2: 100 %
COX15	P-1: 100 % P-2: 100 %	COX20	P-1: 100 % P-2: 100 %	COX4I1	P-1: 100 % P-2: 100 %	COX4I2	P-1: 100 % P-2: 100 %
COX5A	P-1: 100 % P-2: 100 %	COX6A1	P-1: 100 % P-2: 100 %	COX6A2	P-1: 100 % P-2: 100 %	COX6B1	P-1: 100 % P-2: 100 %
COX7B	P-1: 100 % P-2: 100 %	COX8A	P-1: 100 % P-2: 100 %	CP	P-1: 100 % P-2: 100 %	CPT1A	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
CPT2	P-1: 100 % P-2: 100 %	CRAT	P-1: 100 % P-2: 100 %	CRPPA	P-1: 100 % P-2: 100 %	CSGALNACT1	P-1: 100 % P-2: 100 %
CTBP1	P-1: 100 % P-2: 100 %	CTDSP2	P-1: 100 % P-2: 100 %	CTSA	P-1: 100 % P-2: 100 %	CTSC	P-1: 100 % P-2: 100 %
CTSD	P-1: 100 % P-2: 100 %	CTSF	P-1: 100 % P-2: 100 %	CTSK	P-1: 100 % P-2: 100 %	CYC1	P-1: 100 % P-2: 100 %
CYCS	P-1: 100 % P-2: 100 %	CYP24A1	P-1: 100 % P-2: 100 %	CYP27A1	P-1: 100 % P-2: 100 %	CYP7B1	P-1: 100 % P-2: 100 %
D2HGDH	P-1: 100 % P-2: 100 %	DARS2	P-1: 100 % P-2: 100 %	DCC	P-1: 100 % P-2: 100 %	DCDC2	P-1: 100 % P-2: 100 %
DCXR	P-1: 100 % P-2: 100 %	DDOST	P-1: 100 % P-2: 100 %	DDR2	P-1: 100 % P-2: 100 %	DFFB	P-1: 100 % P-2: 100 %
DGUOK	P-1: 100 % P-2: 100 %	DHDDS	P-1: 100 % P-2: 100 %	DIABLO	P-1: 100 % P-2: 100 %	DIAPH1	P-1: 100 % P-2: 100 %
DLAT	P-1: 100 % P-2: 100 %	DLD	P-1: 100 % P-2: 100 %	DLST	P-1: 100 % P-2: 100 %	DNA2	P-1: 100 % P-2: 100 %
DNAJC19	P-1: 100 % P-2: 100 %	DNAJC5	P-1: 100 % P-2: 100 %	DNM1L	P-1: 100 % P-2: 100 %	DOLK	P-1: 100 % P-2: 100 %
DPAGT1	P-1: 100 % P-2: 100 %	DPM1	P-1: 100 % P-2: 100 %	DPM2	P-1: 100 % P-2: 100 %	DPM3	P-1: 100 % P-2: 100 %
DSE	P-1: 100 % P-2: 100 %	DTD1	P-1: 100 % P-2: 100 %	DYM	P-1: 100 % P-2: 100 %	EARS2	P-1: 100 % P-2: 100 %
EBP	P-1: 100 % P-2: 100 %	ECHS1	P-1: 100 % P-2: 100 %	ECSIT	P-1: 100 % P-2: 100 %	EGFL7	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
ELAC2	P-1: 100 % P-2: 100 %	EMC1	P-1: 100 % P-2: 100 %	ENO3	P-1: 100 % P-2: 100 %	EOGT	P-1: 100 % P-2: 100 %
EPM2A	P-1: 100 % P-2: 100 %	ERAL1	P-1: 100 % P-2: 100 %	ETFA	P-1: 100 % P-2: 100 %	ETFB	P-1: 100 % P-2: 100 %
ETFDH	P-1: 100 % P-2: 100 %	ETHE1	P-1: 100 % P-2: 100 %	EXOSC3	P-1: 100 % P-2: 100 %	EXT1	P-1: 100 % P-2: 100 %
EXT2	P-1: 100 % P-2: 100 %	EXTL3	P-1: 100 % P-2: 100 %	FA2H	P-1: 100 % P-2: 100 %	FAH	P-1: 100 % P-2: 100 %
FAM3A	P-1: 100 % P-2: 100 %	FANCA	P-1: 100 % P-2: 100 %	FAR1	P-1: 100 % P-2: 100 %	FARS2	P-1: 100 % P-2: 100 %
FASTKD2	P-1: 100 % P-2: 100 %	FBP1	P-1: 100 % P-2: 100 %	FBXL4	P-1: 100 % P-2: 100 %	FCSK	P-1: 100 % P-2: 100 %
FDXR	P-1: 100 % P-2: 100 %	FGF12	P-1: 100 % P-2: 100 %	FH	P-1: 100 % P-2: 100 %	FKRP	P-1: 100 % P-2: 100 %
FKTN	P-1: 100 % P-2: 100 %	FLAD1	P-1: 100 % P-2: 100 %	FOCAD	P-1: 100 % P-2: 100 %	FOXRED1	P-1: 100 % P-2: 100 %
FUCA1	P-1: 100 % P-2: 100 %	FUT8	P-1: 100 % P-2: 100 %	FXN	P-1: 100 % P-2: 100 %	G6PC2	P-1: 100 % P-2: 100 %
G6PC3	P-1: 100 % P-2: 100 %	G6PD	P-1: 100 % P-2: 100 %	GAA	P-1: 100 % P-2: 100 %	GALC	P-1: 100 % P-2: 100 %
GALE	P-1: 100 % P-2: 100 %	GALK1	P-1: 100 % P-2: 100 %	GALK2	P-1: 100 % P-2: 100 %	GALM	P-1: 100 % P-2: 100 %
GALNS	P-1: 100 % P-2: 100 %	GALNT3	P-1: 100 % P-2: 100 %	GALT	P-1: 100 % P-2: 100 %	GARS1	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
GART	P-1: 100 % P-2: 100 %	GATB	P-1: 100 % P-2: 100 %	GATC	P-1: 100 % P-2: 100 %	GBA	P-1: 100 % P-2: 100 %
GBE1	P-1: 100 % P-2: 100 %	GCK	P-1: 100 % P-2: 100 %	GDAP1	P-1: 100 % P-2: 100 %	GFER	P-1: 100 % P-2: 100 %
GFM1	P-1: 100 % P-2: 100 %	GFM2	P-1: 100 % P-2: 100 %	GFPT1	P-1: 100 % P-2: 100 %	GIMAP5	P-1: 100 % P-2: 100 %
GK	P-1: 100 % P-2: 100 %	GLA	P-1: 100 % P-2: 100 %	GLB1	P-1: 100 % P-2: 100 %	GLRX5	P-1: 100 % P-2: 100 %
GLYAT	P-1: 100 % P-2: 100 %	GM2A	P-1: 100 % P-2: 100 %	GMPPA	P-1: 100 % P-2: 100 %	GMPPB	P-1: 100 % P-2: 100 %
GNE	P-1: 100 % P-2: 100 %	GNPAT	P-1: 100 % P-2: 100 %	GNPDA1	P-1: 100 % P-2: 100 %	GNPTAB	P-1: 100 % P-2: 100 %
GNPTG	P-1: 100 % P-2: 100 %	GNS	P-1: 100 % P-2: 100 %	GORAB	P-1: 100 % P-2: 100 %	GOT2	P-1: 100 % P-2: 100 %
GPI	P-1: 100 % P-2: 100 %	GRN	P-1: 100 % P-2: 100 %	GTPBP3	P-1: 100 % P-2: 100 %	GUF1	P-1: 100 % P-2: 100 %
GUSB	P-1: 100 % P-2: 100 %	GYG1	P-1: 100 % P-2: 100 %	GYG2	P-1: 100 % P-2: 100 %	GYS1	P-1: 100 % P-2: 100 %
GYS2	P-1: 100 % P-2: 100 %	HAAO	P-1: 100 % P-2: 100 %	HADH	P-1: 100 % P-2: 100 %	HADHA	P-1: 100 % P-2: 100 %
HADHB	P-1: 100 % P-2: 100 %	HARS2	P-1: 100 % P-2: 100 %	HCCS	P-1: 100 % P-2: 100 %	HCRT	P-1: 100 % P-2: 100 %
HEXA	P-1: 100 % P-2: 100 %	HEXB	P-1: 98.9 % P-2: 97.6 %	HFE	P-1: 100 % P-2: 100 %	HGSNAT	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
HIBCH	P-1: 100 % P-2: 100 %	HK1	P-1: 100 % P-2: 100 %	HLCS	P-1: 100 % P-2: 100 %	HMGCL	P-1: 100 % P-2: 100 %
HMGCS2	P-1: 100 % P-2: 100 %	HOOK1	P-1: 100 % P-2: 100 %	HSD17B10	P-1: 100 % P-2: 100 %	HSD17B4	P-1: 100 % P-2: 100 %
HSD3B7	P-1: 100 % P-2: 100 %	HSPA9	P-1: 100 % P-2: 100 %	HSPD1	P-1: 100 % P-2: 100 %	HSPE1	P-1: 100 % P-2: 100 %
HTRA2	P-1: 100 % P-2: 100 %	HTT	P-1: 100 % P-2: 100 %	HYAL1	P-1: 100 % P-2: 100 %	IARS2	P-1: 100 % P-2: 100 %
IBA57	P-1: 100 % P-2: 100 %	IDH2	P-1: 100 % P-2: 100 %	IDH3A	P-1: 100 % P-2: 100 %	IDH3B	P-1: 100 % P-2: 100 %
IDS	P-1: 100 % P-2: 100 %	IDUA	P-1: 100 % P-2: 100 %	ISCA1	P-1: 100 % P-2: 100 %	ISCA2	P-1: 100 % P-2: 100 %
ISCU	P-1: 100 % P-2: 100 %	JAG1	P-1: 100 % P-2: 100 %	JAGN1	P-1: 100 % P-2: 100 %	KARS1	P-1: 100 % P-2: 100 %
KCNA4	P-1: 100 % P-2: 100 %	KCTD7	P-1: 100 % P-2: 100 %	KHK	P-1: 100 % P-2: 100 %	KIF12	P-1: 100 % P-2: 100 %
KIF5A	P-1: 100 % P-2: 100 %	KRT17	P-1: 100 % P-2: 100 %	KYNU	P-1: 100 % P-2: 100 %	L2HGDH	P-1: 100 % P-2: 100 %
LAMP2	P-1: 100 % P-2: 100 %	LARGE1	P-1: 100 % P-2: 100 %	LARS2	P-1: 100 % P-2: 100 %	LCT	P-1: 100 % P-2: 100 %
LDHA	P-1: 100 % P-2: 100 %	LDHB	P-1: 100 % P-2: 100 %	LDHD	P-1: 100 % P-2: 100 %	LDLR	P-1: 100 % P-2: 100 %
LFNG	P-1: 100 % P-2: 98.7 %	LIAS	P-1: 100 % P-2: 100 %	LIPA	P-1: 100 % P-2: 100 %	LIPT1	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
LIPT2	P-1: 100 % P-2: 100 %	LONP1	P-1: 100 % P-2: 100 %	LRPPRC	P-1: 100 % P-2: 100 %	LSR	P-1: 100 % P-2: 100 %
LYRM4	P-1: 100 % P-2: 100 %	LYRM7	P-1: 77.2 % P-2: 77.2 %	MAGT1	P-1: 100 % P-2: 100 %	MAN1B1	P-1: 100 % P-2: 100 %
MAN2B1	P-1: 100 % P-2: 100 %	MANBA	P-1: 100 % P-2: 100 %	MAP4K2	P-1: 100 % P-2: 100 %	MARS2	P-1: 100 % P-2: 100 %
MCAT	P-1: 100 % P-2: 100 %	MCOLN1	P-1: 100 % P-2: 100 %	MDH1	P-1: 100 % P-2: 100 %	MDH2	P-1: 100 % P-2: 100 %
MECR	P-1: 100 % P-2: 100 %	MED12	P-1: 100 % P-2: 100 %	MFF	P-1: 100 % P-2: 100 %	MFN2	P-1: 100 % P-2: 100 %
MFSD8	P-1: 100 % P-2: 100 %	MGAT2	P-1: 100 % P-2: 100 %	MGME1	P-1: 100 % P-2: 100 %	MICOS13	P-1: 100 % P-2: 100 %
MICU1	P-1: 100 % P-2: 100 %	MICU2	P-1: 100 % P-2: 100 %	MIEF2	P-1: 100 % P-2: 100 %	MIPEP	P-1: 100 % P-2: 100 %
MOGAT2	P-1: 100 % P-2: 100 %	MOGS	P-1: 100 % P-2: 100 %	MPC1	P-1: 100 % P-2: 100 %	MPDU1	P-1: 100 % P-2: 100 %
MPI	P-1: 100 % P-2: 100 %	MPV17	P-1: 100 % P-2: 100 %	MRM2	P-1: 100 % P-2: 100 %	MRPL12	P-1: 100 % P-2: 100 %
MRPL24	P-1: 100 % P-2: 100 %	MRPL3	P-1: 100 % P-2: 100 %	MRPL44	P-1: 100 % P-2: 100 %	MRPS14	P-1: 100 % P-2: 100 %
MRPS16	P-1: 100 % P-2: 100 %	MRPS2	P-1: 100 % P-2: 100 %	MRPS22	P-1: 100 % P-2: 100 %	MRPS23	P-1: 100 % P-2: 100 %
MRPS25	P-1: 100 % P-2: 100 %	MRPS28	P-1: 100 % P-2: 100 %	MRPS34	P-1: 100 % P-2: 100 %	MRPS35	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
MRPS7	P-1: 100 % P-2: 100 %	MSTO1	P-1: 100 % P-2: 100 %	MTCHI	P-1: 100 % P-2: 100 %	MTFMT	P-1: 100 % P-2: 100 %
MTO1	P-1: 100 % P-2: 100 %	MTPAP	P-1: 100 % P-2: 100 %	MYO5B	P-1: 100 % P-2: 100 %	NADK2	P-1: 100 % P-2: 100 %
NAGA	P-1: 100 % P-2: 100 %	NAGLU	P-1: 100 % P-2: 100 %	NANS	P-1: 100 % P-2: 100 %	NARS2	P-1: 100 % P-2: 100 %
NAT8	P-1: 100 % P-2: 100 %	NAXD	P-1: 100 % P-2: 100 %	NAXE	P-1: 100 % P-2: 100 %	NBAS	P-1: 100 % P-2: 100 %
NDUFA1	P-1: 100 % P-2: 100 %	NDUFA10	P-1: 100 % P-2: 100 %	NDUFA11	P-1: 100 % P-2: 100 %	NDUFA12	P-1: 100 % P-2: 100 %
NDUFA13	P-1: 100 % P-2: 100 %	NDUFA2	P-1: 100 % P-2: 100 %	NDUFA4	P-1: 100 % P-2: 100 %	NDUFA6	P-1: 100 % P-2: 100 %
NDUFA9	P-1: 100 % P-2: 100 %	NDUFAF1	P-1: 100 % P-2: 100 %	NDUFAF2	P-1: 100 % P-2: 100 %	NDUFAF3	P-1: 100 % P-2: 100 %
NDUFAF4	P-1: 100 % P-2: 100 %	NDUFAF5	P-1: 100 % P-2: 100 %	NDUFAF6	P-1: 100 % P-2: 100 %	NDUFAF7	P-1: 100 % P-2: 100 %
NDUFAF8	P-1: 100 % P-2: 100 %	NDUFB10	P-1: 100 % P-2: 100 %	NDUFB11	P-1: 100 % P-2: 100 %	NDUFB3	P-1: 100 % P-2: 100 %
NDUFB8	P-1: 100 % P-2: 100 %	NDUFB9	P-1: 100 % P-2: 100 %	NDUFS1	P-1: 100 % P-2: 100 %	NDUFS2	P-1: 100 % P-2: 100 %
NDUFS3	P-1: 100 % P-2: 100 %	NDUFS4	P-1: 100 % P-2: 100 %	NDUFS6	P-1: 100 % P-2: 100 %	NDUFS7	P-1: 100 % P-2: 100 %
NDUFS8	P-1: 100 % P-2: 100 %	NDUFV1	P-1: 100 % P-2: 100 %	NDUFV2	P-1: 100 % P-2: 100 %	NEU1	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
NFS1	P-1: 100 % P-2: 100 %	NFU1	P-1: 100 % P-2: 100 %	NGLY1	P-1: 100 % P-2: 100 %	NHLRC1	P-1: 100 % P-2: 100 %
NME3	P-1: 100 % P-2: 100 %	NMNAT1	P-1: 100 % P-2: 100 %	NNT	P-1: 100 % P-2: 100 %	NOTCH2	P-1: 100 % P-2: 100 %
NPC1	P-1: 100 % P-2: 100 %	NPC2	P-1: 100 % P-2: 100 %	NRIH4	P-1: 100 % P-2: 100 %	NSUN3	P-1: 100 % P-2: 100 %
NUBPL	P-1: 100 % P-2: 100 %	NUS1	P-1: 100 % P-2: 100 %	OGDH	P-1: 100 % P-2: 100 %	OGT	P-1: 100 % P-2: 100 %
OPA1	P-1: 100 % P-2: 100 %	OPA3	P-1: 100 % P-2: 100 %	OXAIL	P-1: 100 % P-2: 100 %	OXCT1	P-1: 100 % P-2: 100 %
PAM16	P-1: 100 % P-2: 100 %	PANK2	P-1: 100 % P-2: 100 %	PAPSS2	P-1: 100 % P-2: 100 %	PARS2	P-1: 100 % P-2: 100 %
PC	P-1: 100 % P-2: 100 %	PCK1	P-1: 100 % P-2: 100 %	PCK2	P-1: 100 % P-2: 100 %	PDCD6	P-1: 100 % P-2: 100 %
PDE12	P-1: 100 % P-2: 100 %	PDHA1	P-1: 100 % P-2: 100 %	PDHB	P-1: 100 % P-2: 100 %	PDHX	P-1: 100 % P-2: 100 %
PDK3	P-1: 100 % P-2: 100 %	PDP1	P-1: 100 % P-2: 100 %	PDPR	P-1: 100 % P-2: 100 %	PDSS1	P-1: 100 % P-2: 100 %
PDSS2	P-1: 100 % P-2: 100 %	PDZKIPI	P-1: 100 % P-2: 100 %	PET100	P-1: 100 % P-2: 100 %	PETI17	P-1: 100 % P-2: 100 %
PEX1	P-1: 100 % P-2: 100 %	PEX10	P-1: 100 % P-2: 100 %	PEX11A	P-1: 100 % P-2: 100 %	PEX11B	P-1: 100 % P-2: 100 %
PEX12	P-1: 100 % P-2: 100 %	PEX13	P-1: 100 % P-2: 100 %	PEX14	P-1: 100 % P-2: 100 %	PEX16	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
PEX19	P-1: 100 % P-2: 100 %	PEX2	P-1: 100 % P-2: 100 %	PEX26	P-1: 100 % P-2: 100 %	PEX3	P-1: 100 % P-2: 100 %
PEX5	P-1: 100 % P-2: 100 %	PEX6	P-1: 100 % P-2: 100 %	PEX7	P-1: 100 % P-2: 100 %	PFKM	P-1: 100 % P-2: 100 %
PGAM2	P-1: 100 % P-2: 100 %	PGAP1	P-1: 95.6 % P-2: 100 %	PGAP2	P-1: 100 % P-2: 100 %	PGAP3	P-1: 100 % P-2: 100 %
PGK1	P-1: 100 % P-2: 100 %	PGM1	P-1: 100 % P-2: 100 %	PGM3	P-1: 100 % P-2: 100 %	PHKA1	P-1: 100 % P-2: 100 %
PHKA2	P-1: 100 % P-2: 100 %	PHKB	P-1: 100 % P-2: 100 %	PHKG2	P-1: 100 % P-2: 100 %	PHYH	P-1: 100 % P-2: 100 %
PIGA	P-1: 100 % P-2: 100 %	PIGB	P-1: 100 % P-2: 100 %	PIGC	P-1: 100 % P-2: 100 %	PIGG	P-1: 100 % P-2: 100 %
PIGL	P-1: 100 % P-2: 100 %	PIGM	P-1: 100 % P-2: 100 %	PIGN	P-1: 100 % P-2: 100 %	PIGO	P-1: 100 % P-2: 100 %
PIGP	P-1: 100 % P-2: 100 %	PIGQ	P-1: 100 % P-2: 100 %	PIGT	P-1: 100 % P-2: 100 %	PIGU	P-1: 100 % P-2: 100 %
PIGV	P-1: 100 % P-2: 100 %	PIGW	P-1: 100 % P-2: 100 %	PIGY	P-1: 100 % P-2: 100 %	PINK1	P-1: 100 % P-2: 100 %
PISD	P-1: 100 % P-2: 100 %	PITRM1	P-1: 100 % P-2: 100 %	PKHD1	P-1: 100 % P-2: 100 %	PKLR	P-1: 100 % P-2: 100 %
PLA2G6	P-1: 100 % P-2: 100 %	PLPP6	P-1: 100 % P-2: 100 %	PMEL	P-1: 100 % P-2: 100 %	PMM2	P-1: 100 % P-2: 100 %
PMPCA	P-1: 100 % P-2: 100 %	PMPCB	P-1: 100 % P-2: 100 %	PNPLA4	P-1: 100 % P-2: 100 %	PNPLA8	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
PNPT1	P-1: 100 % P-2: 100 %	PODXL	P-1: 100 % P-2: 99.6 %	POFUT1	P-1: 100 % P-2: 100 %	POGLUT1	P-1: 100 % P-2: 100 %
POLG	P-1: 100 % P-2: 100 %	POLG2	P-1: 100 % P-2: 100 %	POLRMT	P-1: 100 % P-2: 100 %	POMGNT1	P-1: 100 % P-2: 100 %
POMGNT2	P-1: 100 % P-2: 100 %	POMK	P-1: 100 % P-2: 100 %	POMT1	P-1: 100 % P-2: 100 %	POMT2	P-1: 100 % P-2: 100 %
POPI	P-1: 100 % P-2: 100 %	PPA2	P-1: 96.3 % P-2: 96.3 %	PPCS	P-1: 100 % P-2: 100 %	PPOX	P-1: 100 % P-2: 100 %
PPT1	P-1: 100 % P-2: 100 %	PRKAG2	P-1: 91.9 % P-2: 96.8 %	PRKAG3	P-1: 100 % P-2: 100 %	PRKN	P-1: 100 % P-2: 100 %
PROC	P-1: 100 % P-2: 100 %	PSAP	P-1: 100 % P-2: 100 %	PTCD3	P-1: 100 % P-2: 100 %	PTRH2	P-1: 100 % P-2: 100 %
PUS1	P-1: 100 % P-2: 100 %	PYCR1	P-1: 100 % P-2: 100 %	PYDC1	P-1: 100 % P-2: 100 %	PYGL	P-1: 100 % P-2: 100 %
PYGM	P-1: 100 % P-2: 100 %	QRSL1	P-1: 100 % P-2: 100 %	RARS2	P-1: 100 % P-2: 100 %	RBCK1	P-1: 100 % P-2: 100 %
RFT1	P-1: 100 % P-2: 100 %	RINT1	P-1: 100 % P-2: 100 %	RMND1	P-1: 100 % P-2: 100 %	RNASEH1	P-1: 100 % P-2: 100 %
ROBO3	P-1: 100 % P-2: 100 %	RPIA	P-1: 100 % P-2: 100 %	RPN2	P-1: 100 % P-2: 100 %	RRM2B	P-1: 100 % P-2: 100 %
RTN4IP1	P-1: 100 % P-2: 100 %	RXYLT1	P-1: 100 % P-2: 100 %	SACS	P-1: 100 % P-2: 100 %	SAMHD1	P-1: 100 % P-2: 100 %
SARS2	P-1: 100 % P-2: 100 %	SCGBID1	P-1: 100 % P-2: 100 %	SCO1	P-1: 100 % P-2: 100 %	SCO2	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
SCP2	P-1: 100 % P-2: 100 %	SDHA	P-1: 100 % P-2: 100 %	SDHAF1	P-1: 100 % P-2: 100 %	SDHAF2	P-1: 100 % P-2: 100 %
SDHB	P-1: 100 % P-2: 100 %	SDHC	P-1: 100 % P-2: 100 %	SDHD	P-1: 100 % P-2: 100 %	SEC23A	P-1: 100 % P-2: 100 %
SEC23B	P-1: 100 % P-2: 100 %	SECISBP2	P-1: 100 % P-2: 100 %	SEPSECS	P-1: 100 % P-2: 100 %	SERAC1	P-1: 100 % P-2: 100 %
SERPINA1	P-1: 100 % P-2: 100 %	SFII	P-1: 100 % P-2: 100 %	SFTPA1	P-1: 100 % P-2: 100 %	SFTPA2	P-1: 100 % P-2: 100 %
SFXN4	P-1: 100 % P-2: 100 %	SGSH	P-1: 100 % P-2: 100 %	SH3D19	P-1: 100 % P-2: 100 %	SHPK	P-1: 100 % P-2: 100 %
SI	P-1: 98.2 % P-2: 98.2 %	SLC10A7	P-1: 100 % P-2: 100 %	SLC17A5	P-1: 100 % P-2: 100 %	SLC19A2	P-1: 100 % P-2: 100 %
SLC19A3	P-1: 100 % P-2: 100 %	SLC22A5	P-1: 100 % P-2: 100 %	SLC25A1	P-1: 100 % P-2: 100 %	SLC25A10	P-1: 100 % P-2: 100 %
SLC25A11	P-1: 100 % P-2: 100 %	SLC25A12	P-1: 100 % P-2: 100 %	SLC25A13	P-1: 100 % P-2: 100 %	SLC25A15	P-1: 100 % P-2: 100 %
SLC25A19	P-1: 100 % P-2: 100 %	SLC25A20	P-1: 100 % P-2: 100 %	SLC25A21	P-1: 100 % P-2: 100 %	SLC25A22	P-1: 100 % P-2: 100 %
SLC25A24	P-1: 100 % P-2: 100 %	SLC25A26	P-1: 100 % P-2: 100 %	SLC25A3	P-1: 100 % P-2: 100 %	SLC25A32	P-1: 100 % P-2: 100 %
SLC25A38	P-1: 100 % P-2: 100 %	SLC25A4	P-1: 100 % P-2: 100 %	SLC25A42	P-1: 100 % P-2: 100 %	SLC25A46	P-1: 100 % P-2: 100 %
SLC26A2	P-1: 100 % P-2: 100 %	SLC2A1	P-1: 100 % P-2: 100 %	SLC2A2	P-1: 100 % P-2: 100 %	SLC33A1	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
SLC35A1	P-1: 100 % P-2: 100 %	SLC35A2	P-1: 100 % P-2: 100 %	SLC35A3	P-1: 100 % P-2: 100 %	SLC35C1	P-1: 100 % P-2: 100 %
SLC35D1	P-1: 92.2 % P-2: 92.2 %	SLC37A4	P-1: 99.9 % P-2: 99.9 %	SLC39A8	P-1: 100 % P-2: 100 %	SLC52A1	P-1: 100 % P-2: 100 %
SLC52A2	P-1: 100 % P-2: 100 %	SLC52A3	P-1: 100 % P-2: 100 %	SLC5A1	P-1: 100 % P-2: 100 %	SLC5A2	P-1: 100 % P-2: 100 %
SLC6A4	P-1: 100 % P-2: 100 %	SLC9A7	P-1: 100 % P-2: 100 %	SLPI	P-1: 100 % P-2: 100 %	SMPD1	P-1: 100 % P-2: 100 %
SOAT1	P-1: 100 % P-2: 100 %	SORD	P-1: 100 % P-2: 100 %	SP8	P-1: 100 % P-2: 100 %	SPAM1	P-1: 100 % P-2: 100 %
SPART	P-1: 100 % P-2: 100 %	SPATA5	P-1: 100 % P-2: 100 %	SPG7	P-1: 100 % P-2: 100 %	SQOR	P-1: 100 % P-2: 100 %
SRD5A3	P-1: 100 % P-2: 100 %	SSBP1	P-1: 100 % P-2: 100 %	SSR3	P-1: 100 % P-2: 100 %	SSR4	P-1: 100 % P-2: 100 %
ST3GAL3	P-1: 100 % P-2: 100 %	ST3GAL5	P-1: 100 % P-2: 100 %	STAT2	P-1: 100 % P-2: 100 %	STS	P-1: 100 % P-2: 100 %
STT3A	P-1: 100 % P-2: 100 %	STT3B	P-1: 100 % P-2: 100 %	STXBPI	P-1: 100 % P-2: 100 %	SUCLA2	P-1: 100 % P-2: 100 %
SUCLG1	P-1: 100 % P-2: 100 %	SUGCT	P-1: 100 % P-2: 100 %	SUMF1	P-1: 100 % P-2: 100 %	SURF1	P-1: 100 % P-2: 100 %
SYCP2	P-1: 97.2 % P-2: 97.2 %	TAC4	P-1: 100 % P-2: 100 %	TACO1	P-1: 100 % P-2: 100 %	TALDO1	P-1: 100 % P-2: 100 %
TANGO2	P-1: 100 % P-2: 100 %	TARS2	P-1: 100 % P-2: 100 %	TFAM	P-1: 100 % P-2: 100 %	TGFBRAPI	P-1: 100 % P-2: 100 %

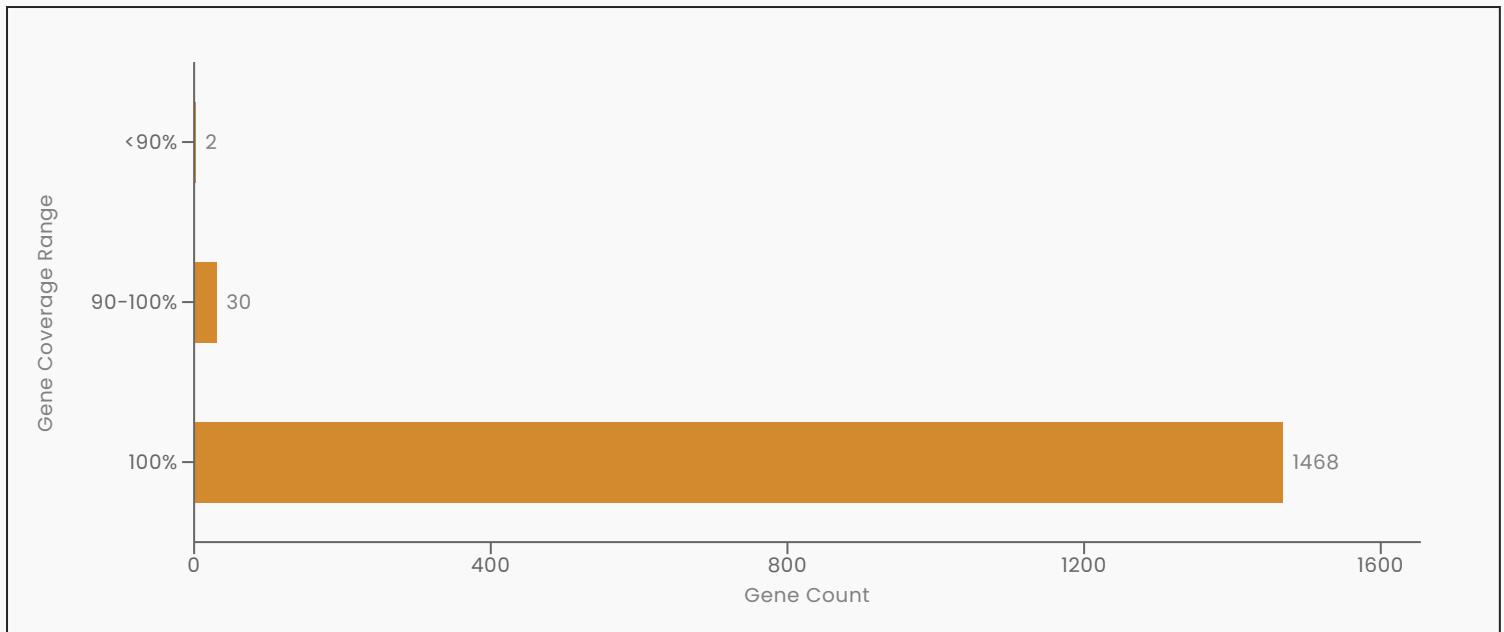
Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
THGIL	P-1: 100 % P-2: 100 %	TIMM22	P-1: 100 % P-2: 100 %	TIMM50	P-1: 100 % P-2: 100 %	TIMM8A	P-1: 100 % P-2: 100 %
TIMMDC1	P-1: 100 % P-2: 100 %	TJP2	P-1: 100 % P-2: 100 %	TK1	P-1: 100 % P-2: 100 %	TK2	P-1: 100 % P-2: 100 %
TKFC	P-1: 100 % P-2: 100 %	TKT	P-1: 100 % P-2: 100 %	TMEM126A	P-1: 100 % P-2: 100 %	TMEM126B	P-1: 100 % P-2: 100 %
TMEM165	P-1: 100 % P-2: 100 %	TMEM199	P-1: 100 % P-2: 100 %	TMEM65	P-1: 92.4 % P-2: 92.4 %	TMEM70	P-1: 100 % P-2: 100 %
TOMM70	P-1: 100 % P-2: 100 %	TOP3A	P-1: 100 % P-2: 100 %	TP53INP2	P-1: 100 % P-2: 100 %	TPI1	P-1: 100 % P-2: 100 %
TPK1	P-1: 100 % P-2: 100 %	TPP1	P-1: 100 % P-2: 100 %	TRAK1	P-1: 100 % P-2: 100 %	TRAP1	P-1: 100 % P-2: 100 %
TRAPPC11	P-1: 100 % P-2: 100 %	TRAPPC12	P-1: 100 % P-2: 100 %	TRAPPC6B	P-1: 100 % P-2: 100 %	TRAPPC9	P-1: 100 % P-2: 100 %
TREH	P-1: 100 % P-2: 100 %	TRIM37	P-1: 100 % P-2: 100 %	TRIP11	P-1: 100 % P-2: 100 %	TRIT1	P-1: 100 % P-2: 100 %
TRMT10C	P-1: 100 % P-2: 100 %	TRMT5	P-1: 100 % P-2: 100 %	TRMU	P-1: 100 % P-2: 100 %	TRNT1	P-1: 100 % P-2: 100 %
TSFM	P-1: 100 % P-2: 100 %	TTC19	P-1: 100 % P-2: 100 %	TUFM	P-1: 100 % P-2: 100 %	TULP3	P-1: 100 % P-2: 100 %
TUSC3	P-1: 100 % P-2: 100 %	TWNK	P-1: 100 % P-2: 100 %	TXN2	P-1: 100 % P-2: 100 %	TXNIP	P-1: 100 % P-2: 100 %
TYMP	P-1: 100 % P-2: 100 %	UBL4A	P-1: 100 % P-2: 100 %	UGT1A1	P-1: 100 % P-2: 100 %	UQCC2	P-1: 100 % P-2: 100 %

Gene Coverage

Gene	Coverage	Gene	Coverage	Gene	Coverage	Gene	Coverage
UQCC3	P-1: 100 % P-2: 100 %	UQCRB	P-1: 100 % P-2: 100 %	UQCRC2	P-1: 100 % P-2: 100 %	UQCRFS1	P-1: 100 % P-2: 100 %
UQCRQ	P-1: 100 % P-2: 100 %	USP53	P-1: 100 % P-2: 100 %	VAR1	P-1: 100 % P-2: 100 %	VAR2	P-1: 100 % P-2: 100 %
VIPAS39	P-1: 100 % P-2: 100 %	VMA21	P-1: 100 % P-2: 100 %	VPS13B	P-1: 100 % P-2: 100 %	VPS13C	P-1: 99.1 % P-2: 99.6 %
VPS33A	P-1: 100 % P-2: 100 %	VPS33B	P-1: 100 % P-2: 100 %	VPS50	P-1: 100 % P-2: 100 %	WARS2	P-1: 100 % P-2: 100 %
WFS1	P-1: 100 % P-2: 100 %	WWTR1	P-1: 100 % P-2: 100 %	XPNPEP3	P-1: 100 % P-2: 100 %	XRCC4	P-1: 100 % P-2: 100 %
XYLT1	P-1: 100 % P-2: 100 %	XYLT2	P-1: 100 % P-2: 100 %	YARS2	P-1: 100 % P-2: 100 %	YME1L1	P-1: 100 % P-2: 100 %
ZFYVE19	P-1: 100 % P-2: 100 %	ZNF260	P-1: 100 % P-2: 100 %				

Gene Coverage Distribution



QC Metrics

	Partner 1	Partner 2
Total aligned reads	99.94 %	99.94 %
Total reads	137.00 (M)	95.49 (M)
Total data generated	20.43 (Gb)	14.17 (Gb)
Total reads which passed mapping quality cutt-off	19.72 (Gb)	13.64 (Gb)

Reviewed By

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